Unnamed_Unit

Unit 1 - Conditional Statements

PDF Documents

PDF Document 1: 1734175205136-Chapter 3 - unit 1 - Conditional Statements.pdf

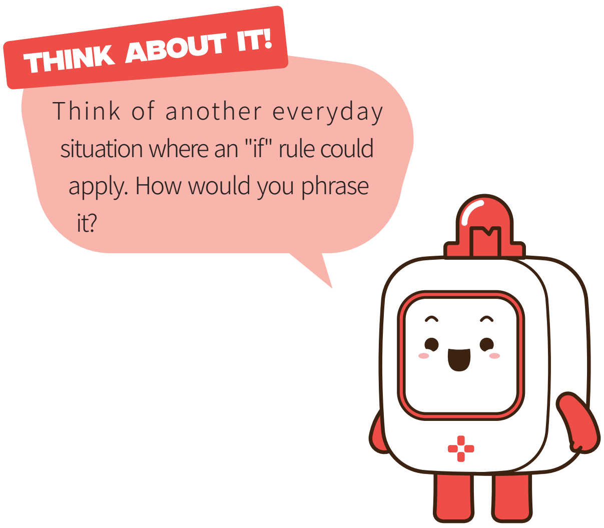
This PDF document is included in the unit materials.

UNIT 1 CONDITIONAL STATEMENTS

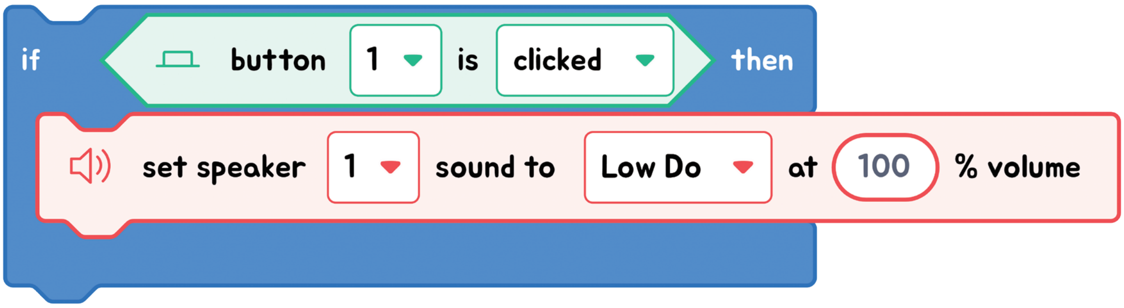
If Statement

Imagine you're deciding whether to carry an umbrella outside. The simple rule you might follow is: "If it's raining, then I'll take an umbrella. If it's not raining, I won't need one."

This decision-making process is similar to how an "if statement" functions in programming. It acts like a rule or condition, essentially saying, "If a certain condition is true, then perform a specific action."



EXPLANATION WITH CODE EXAMPLE



In block coding, we use the "if statement" to check conditions. If the condition is met (or true), the program will carry out a specific task. The "if statement" block visually separates the condition from the action with "if" and "then" parts. It examines the condition, and if the condition holds true, it proceeds to execute the task inside the block.

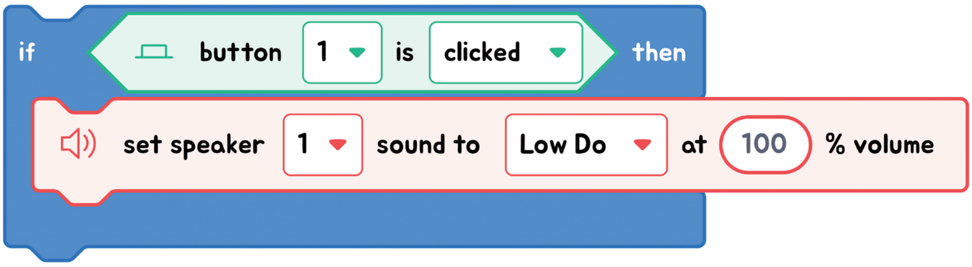
Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the tasks inside the "if" statement are carried out. These tasks specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."

Exercise

Finding Conditions and Tasks:
Look at the block example and identify the condition it checks and the task it performs!

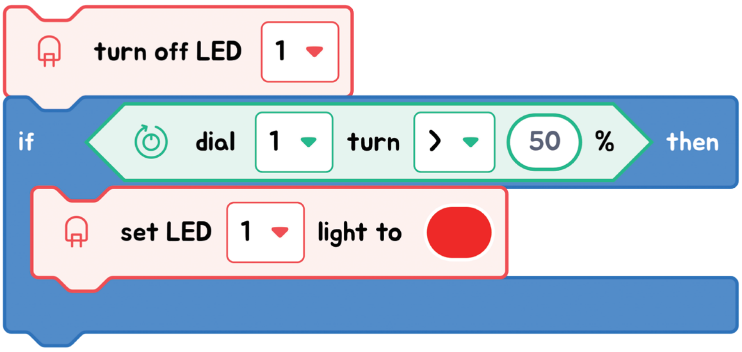
Exercise 1



Condition(s)

Task(s)

Exercise 2



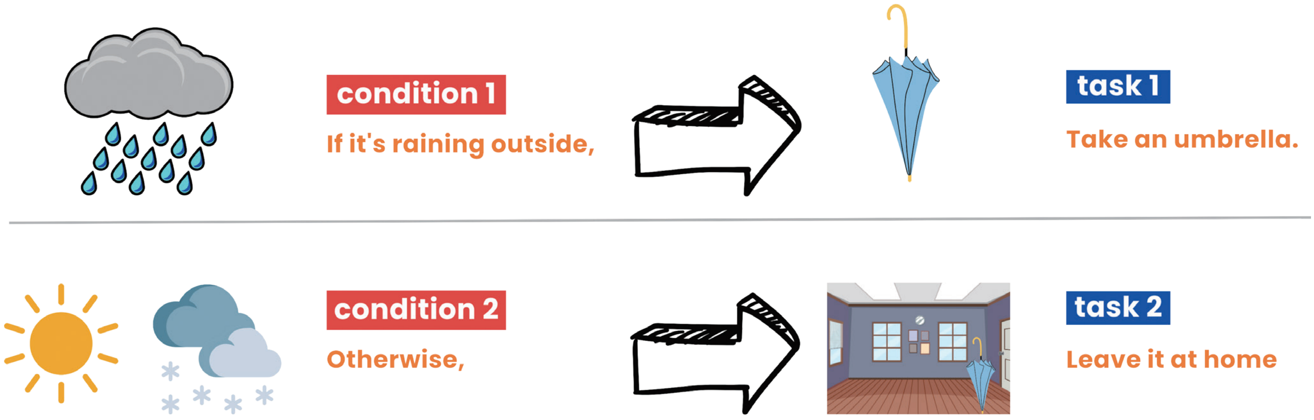
Condition(s)

Task(s)

If else Statement

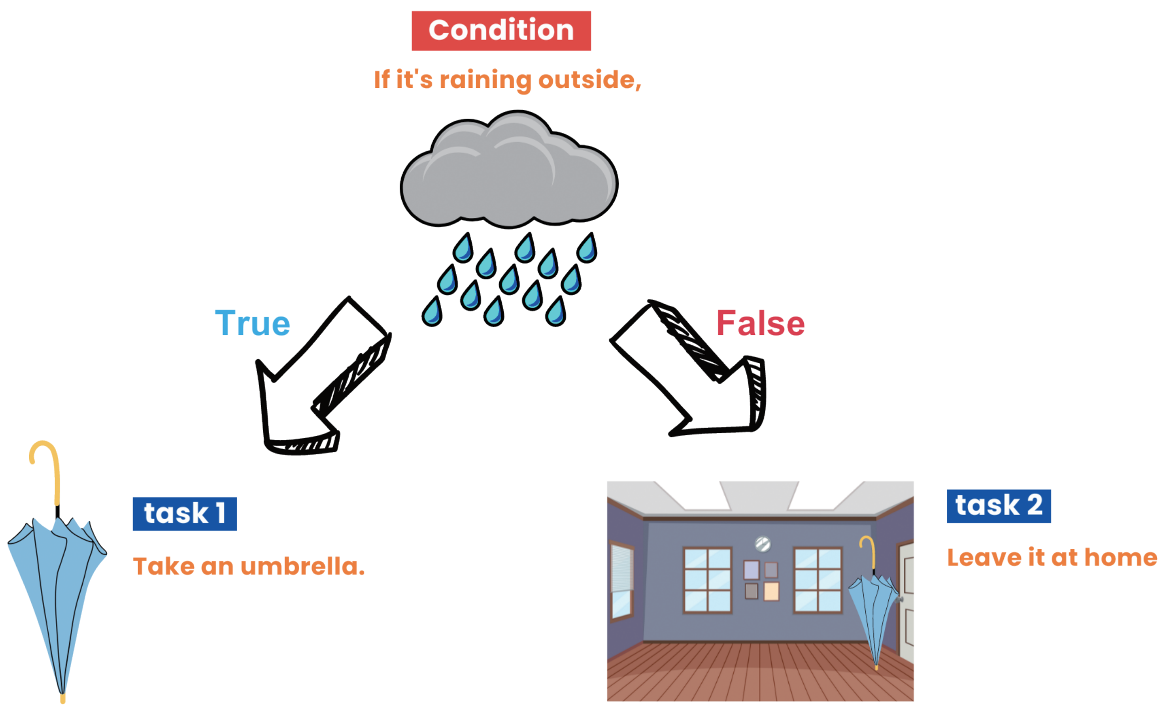
An 'If-Else' statement is like an 'If' statement with a backup plan. If the 'If' part isn't true, then the 'Else' part takes over and does something else instead.

✓ ELSE



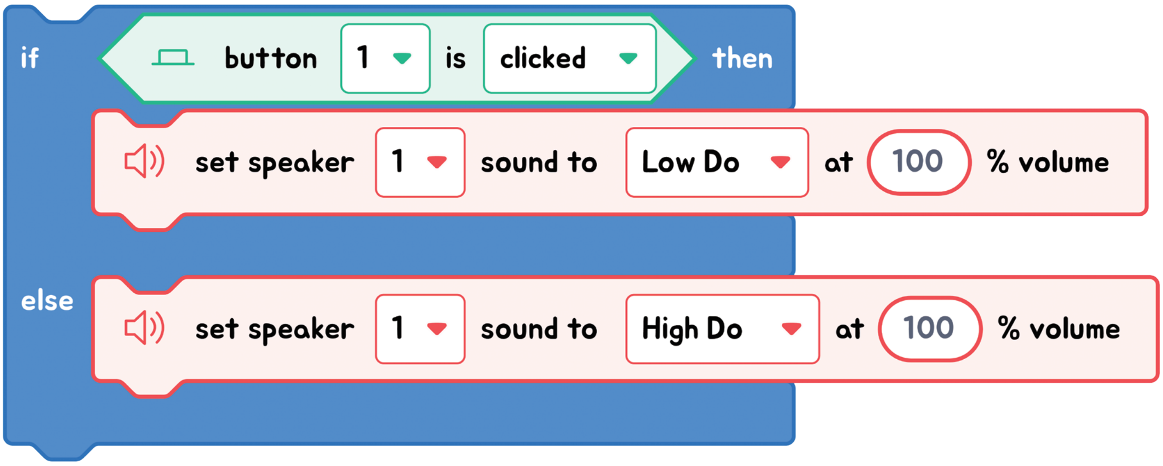
Imagine you're deciding whether to bring an umbrella. An 'If-Else' statement can help you with this!

If it's raining outside (this is the condition), you'll take an umbrella (this is the task). Otherwise (this is the Else part), you'll leave the umbrella at home. So, even though we only consider rain, Else takes care of everything else by default.

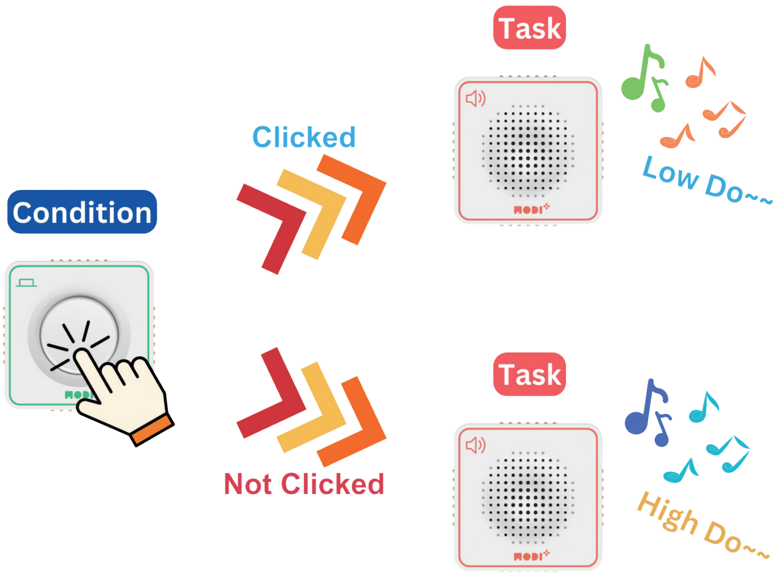


The key to 'If-Else' is that if the 'If' condition isn't met, the program will do what's in the 'Else' part instead.

EXPLANATION WITH CODE EXAMPLE



In block coding, the "If-Else" statement lets us choose between two actions based on a condition. If the condition is true, one action is taken; if not, a different action is executed. The "If-Else" block visually splits this choice, ensuring the program always has a clear direction, no matter the condition's outcome.



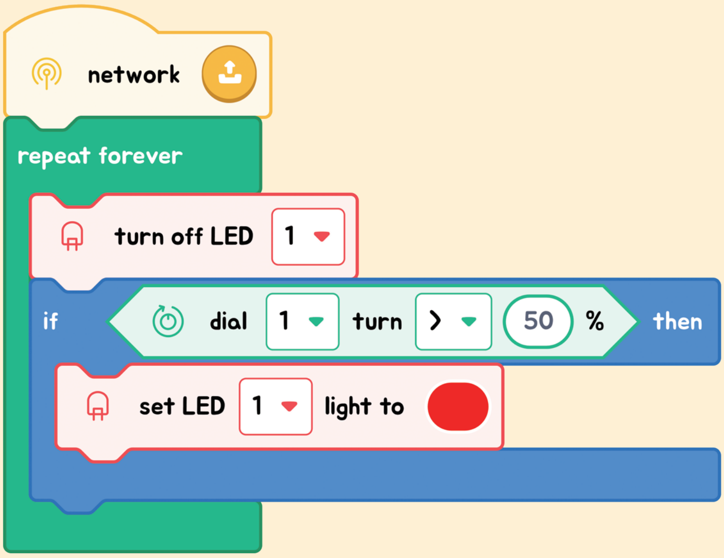
Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the instructions inside the "if" statement are carried out. These instructions specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."
3. **Execution if False:** If the condition is not met, then the instructions inside the "else" statement are carried out. These instructions specify the actions to take when the condition is not met. Thus, the task performed is "Set the speaker to play the 'high do' sound at 100% volume."

Exercise

Exercise 1

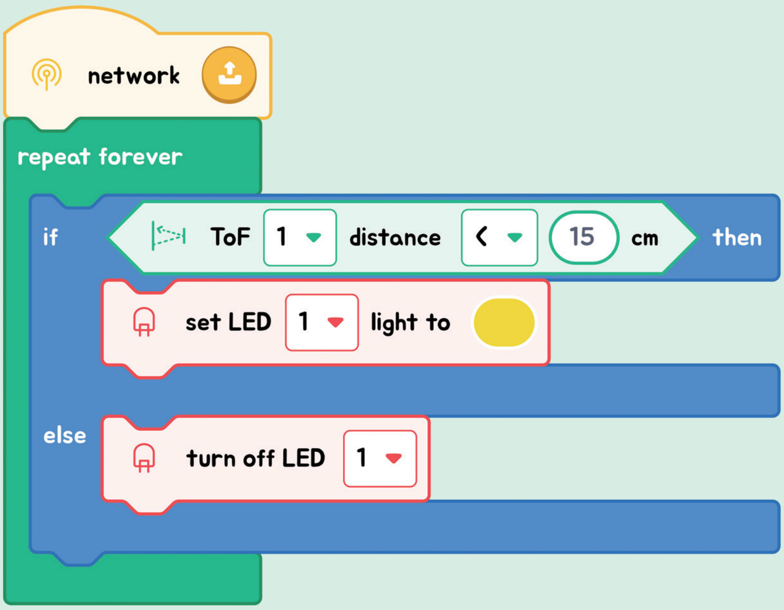
Change this code to an **'If-Else' statement** in the Code Editor.



Exercise 2

Use the Code Editor to create the following project:

"We're making a sensor light. If an object is close to the sensor, the light will turn on. If there's no object nearby, the LED will be off."



TIPS

By monitoring with the ToF (Time of Flight) sensor, determine the distance at which you want the light to turn on.